

REQUEST FOR QUOTATION

Closed-Loop Cooling Water System — Mill Process Equipment

Ref: SAWIT-RFQ-2026-014 | Date: 2 June 2026 | Closing: 25 June 2026

1. PROJECT BACKGROUND

Sawit Maju Sdn Bhd is upgrading the cooling water utility system at our palm oil mill in Pengerang, Johor (60 tonnes FFB/hr capacity). The current open-loop cooling tower configuration has reached end-of-life and we are transitioning to a closed-loop chilled water system to serve our sterilisation autoclave heat exchangers, kernel oil refinery condensers, and ancillary process cooling loads.

This project must be commissioned before our annual major maintenance shutdown in Q4 2026.

2. SCOPE OF SUPPLY & INSTALLATION

The contractor shall supply, install, test, and commission a complete cooling water circulation system including:

- Duty + standby centrifugal pumps with VFD control (one duty, one standby, automatic transfer on fault)
- Suction header, discharge header, and expansion vessel — closed loop configuration
- Full instrumentation: flow transmitter on main line, pressure transmitters on pump suction and discharge, temperature transmitters on supply and return lines, level transmitter on expansion vessel
- Pressure safety relief valves (PSV) — DOSH-registered
- Control valve for temperature regulation, all isolation, check, drain, and vent valves
- Carbon steel piping (epoxy-lined for corrosion resistance in Malaysian tropical climate)
- MCC panel with VFD drives, local control panel with HMI, all power and instrument cabling
- Equipment skid, pipe supports, and access platforms
- Y-strainer (commissioning), flushing services, performance testing

3. TECHNICAL REQUIREMENTS

Fluid	Cooling water (closed loop, treated)
Design flow rate	Approximately 80 m³/hr (size to actual heat load)
Operating pressure	6 bar(g) typical
Operating temperature	32°C supply / 42°C return (typical)
Duty cycle	24 hours/day, 7 days/week, continuous operation
Service life	Minimum 15 years under tropical climate
Redundancy	N+1 pump redundancy with automatic switchover

4. COMPLIANCE & STANDARDS

All equipment and installation shall conform to:

- DOSH FMA 1967 — pressure vessel registration and rotating equipment certification
- BOMBA UBBL — fire safety for electrical enclosures in industrial buildings
- SIRIM QAS — electrical equipment certification (VFD, control panels)
- DOE Malaysia — environmental compliance (cooling water discharge)
- MS 1745 — code of practice for installation of metallic piping systems

5. COMMERCIAL TERMS

Project budget ceiling: RM 750,000 (Malaysian Ringgit, inclusive of SST). All pricing in MYR. Payment terms: 30% on order, 50% on delivery, 20% on commissioning sign-off.

Preferred suppliers: Malaysian-registered companies (Sdn Bhd) with offices in Johor, Selangor, or Penang for after-sales support proximity to Pengerang.

Lead time requirement: equipment delivery within 10 weeks of purchase order, installation and commissioning within a further 6 weeks.

SUBMIT TO:

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